Solving Rational Equations

Name _____

Algebra 2 Lesson 4-5 (fast track) Date _____ Period _____

Learning goal: I can solve a rational equation, including a work-rate problem, by multiplying both sides by a common denominator, and I can identify any extraneous solutions.

Language goal: I can explain why a solution is or is not valid using the words rational equation, common denominator, rate, domain and extraneous solution.

Spanish: *ecuación racional* (rational equation), *solución extraña* (extraneous solution), *dominio* (domain).

On the quiz: (2) graded: add the workers' **rates** (Example 2, Practice 1). (4) graded: clear the fractions, solve, cross out the **extraneous solution** (Example 4). (3) bonus: a variable denominator on each side (Practice 2, 3). (1), (5): full Savvas quiz only.

On the Topic 4 test: Questions 7 and 13 ask for a time alone or together (Example 2). Questions 9 and 11 ask you to solve and check (Example 4). Questions 6 and 17 (a denominator like $x(x-3)$) come from Example 3, which this packet skips; Practice 2 rehearses them.

Example 2: Solve a Work-Rate Problem

Arthur and Cheyenne are making Chinese paper cutting designs to give as gifts for the coming year. Together they can complete the job in 24 hours. Cheyenne works twice as fast as Arthur. How long would it take Cheyenne to complete the job if she was working alone?

Step 1 Determine the work-rates. The job done per hour is the **rate**. Let x = Arthur's hours alone.

Arthur: $\frac{1 \text{ job}}{x \text{ hours}} = \frac{1}{x}$ job per hour.

Cheyenne: $\frac{1 \text{ job}}{\frac{1}{2}x \text{ hours}} = \frac{2}{x}$ job per hour.

Together: $\frac{1 \text{ job}}{24 \text{ hours}} = \frac{1}{24}$ job per hour.

Watch out: twice as fast means *half* the time, $\frac{1}{2}x$, not $2x$.

Step 2 Write the **rational equation** for their rates together. Solve. Domain: $x \neq 0$.

$$\frac{1}{x} + \frac{2}{x} = \frac{1}{24}$$

$$24x\left(\frac{1}{x} + \frac{2}{x}\right) = \left(\frac{1}{24}\right)24x$$

$$24 + 48 = x \quad \Rightarrow \quad x = 72$$

Check: $\frac{1}{72} + \frac{2}{72} = \frac{3}{72} = \frac{1}{24}$ ✓

Arthur: 72 hours. Cheyenne alone: 36 hours.

Say it: Arthur's _____ is $\frac{1}{x}$ job per hour. I multiplied both sides of the _____ by $24x$, the _____. The answer 72 is in the _____.

Try It 2: Solve a work-rate problem

It takes 12 hours to fill a pool with two pipes, where the water in one pipe flows three times as fast as the other pipe. How long will it take the slower pipe to fill the pool by itself?

Hint: x = slower pipe's hours alone. Rates: slower _____, faster _____. Equation: _____

The slower pipe alone: _____ hours

Example 4: Check for Extraneous Solutions

What are the solutions to the following equations? First find the **domain**: the values that make a denominator 0 can never be solutions. A value that the cleared equation gives but the domain leaves out is an **extraneous solution**.

A. $\dfrac{5x}{x-2} = 7 + \dfrac{10}{x-2}$

Domain: $x \neq 2$.

Common denominator: $x - 2$.

$$(x-2)\left(\frac{5x}{x-2}\right) = \left(7 + \frac{10}{x-2}\right)(x-2)$$
$$5x = 7(x-2) + 10$$
$$5x = 7x - 14 + 10$$
$$-2x = -4$$
$$x = 2$$

Check: $x = 2$ makes the denominator $x - 2$ equal to 0, so 2 is an **extraneous solution**.

This equation has **no solution**.

B. $\dfrac{3}{x-3} = \dfrac{x}{x-3} - \dfrac{x}{4}$

Domain: $x \neq 3$. Common denominator: $4(x-3)$.

$$4(x-3)\left(\frac{3}{x-3}\right) = \left(\frac{x}{x-3} - \frac{x}{4}\right)4(x-3)$$
$$4(3) = 4(x) - x(x-3)$$
$$12 = 4x - x^2 + 3x$$
$$x^2 - 7x + 12 = 0 \quad \text{standard form}$$
$$(x-3)(x-4) = 0 \quad \text{factor}$$
$$x = 3 \quad \text{or} \quad x = 4$$

Check: $x = 3$ makes $x - 3 = 0$, so 3 is extraneous.

$x = 4$: $\dfrac{3}{1} = 3$ and $\dfrac{4}{1} - \dfrac{4}{4} = 4 - 1 = 3$ ✓

The only solution is $x = 4$; 3 is extraneous.

Say it: In Part B, $x = 3$ is an _____ because it makes the _____ equal to 0, so it is not in the _____ of the original equation. The only solution is _____.

Try It 4: What are the solutions to the following equations?

a. $x + \dfrac{6}{x-3} = \dfrac{2x}{x-3}$

Domain: $x \neq$ _____ Multiply by: _____

Solutions of the cleared equation: _____

Extraneous: _____ Solution: _____

b. $\dfrac{x^2}{x+5} = \dfrac{25}{x+5}$

Domain: $x \neq$ _____ Multiply by: _____

Solutions of the cleared equation: _____

Extraneous: _____ Solution: _____

In part b, the cleared equation has two solutions. Why is only one of them a solution of the original equation? Use the word *extraneous*.

Practice: Working together, then two denominators

1. Working together (Practice 26). Kenji can finish a puzzle in 2 hours working alone. Oscar can finish the same puzzle in 3 hours working alone. How long would it take Oscar and Kenji to finish the puzzle if they worked on it together?

a. Each rate is 1 puzzle divided by the hours. Kenji: _____ puzzle per hour. Oscar: _____ puzzle per hour.

b. Let t = hours working together. Their rates add up to $\frac{1}{t}$: _____ + _____ = _____

c. Multiply both sides by the common denominator _____ and solve.

d. $t =$ _____ hours

e. (Practice 8) If you solve a work-rate problem and your solution, which represents the amount of time it would take working together, exceeds the individual working alone times that are given, then how do you know your solution is unreasonable? Explain.

2. A product denominator (Practice 24). Solve the equation. $\dfrac{x}{x-3}+\dfrac{2x}{x+3}=\dfrac{18}{(x+3)(x-3)}$

Domain: $x \neq$ _____ and _____ Multiply both sides by: _____

Solutions of the cleared equation: _____ Extraneous: _____ Solution: _____

3. Two denominators (Practice 32). Which of the following is the solution of $\dfrac{3}{x+1}=\dfrac{2}{x-3}$?

A. $x=-11$ B. $x=-\frac{7}{5}$ C. $x=\frac{7}{5}$ D. $x=11$

Multiply both sides by *both* denominators: _____

Solution: $x =$ _____ Check: ✓ or ×

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
CHP ★	Solve rational equations	Quiz 2 (the rate equation), 3, 4; Quiz 1 (full Savvas quiz only)
CJC ★	Solve a quadratic equation by factoring	Quiz 4: the quadratic left after clearing the fractions (Example 4B)
7JJ	Rational functions: asymptotes and excluded values	Warm-up: the excluded values are the numbers that can never be solutions
FEX	Add and subtract rational expressions	Warm-up from 4-4: finding a common denominator

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Practice 13 (DOK 3)

Higher Order Thinking. Write a rational equation that cannot have 2 or -6 as solutions.

A denominator that is 0 when $x = 2$: _____ and when $x = -6$: _____

My equation:

_____

_____

Why can neither 2 nor -6 be a solution of your equation? Use the words *domain* and *denominator*.

_____

_____

_____

Before the quiz, finish these sentences.

To solve a rational equation, I first find the _____, then I multiply both sides by the _____.

After I solve, I check each answer. An answer that makes a denominator 0 is an _____, so I cross it out.

When two people work together, I _____ their _____, and the time together is _____ than either time alone.